

XAVIER ROYER

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EDUCATION

California Polytechnic State University, San Luis Obispo, B.S. Physics and Computer Science Expected May 2029
• GPA: 3.973 | Cal Poly Distance Club | Hack4Impact

SKILLS

Programming/tools: Python, C, Java, Jupyter, Matlab, TypeScript, React, MongoDB, Godot, GitHub, APIs
Documents/data: Excel, Google Sheets, One Drive, Google Drive
Languages: English and French

TECHNICAL EXPERIENCE

Fish Tracking and Modelling Research, Cal Poly 2026 - Present
• Collaborate with Dr. Tabatabai to track and predict fish movement using neural networks.
• Collect three-dimensional movement data using DLTdv tracking software.

FROST Summer Research Program, Cal Poly Summer 2026
• Estimated stellar-cluster ages from main-sequence turnoff data with Dr. Jeffery.
• Used remote servers, Jupyter Notebook, MATLAB, and databases to run Bayesian inference and analyze data.

Hack4Impact - Meals That Connect, Cal Poly 2025-2026
• Contributed to a student-built website for a local nonprofit's recipe and menu project.
• Used React, MongoDB, GitHub, and Google Drive for development tasks and project coordination.

Programming Projects, Independent / client projects 2024-2025
• Built a circular calendar app in Godot for a client, with circular and linear views, categories, recurring tasks, and saved entries
• Produced UML diagrams, flowcharts, success criteria, test notes, and a user manual to document project delivery.
• Worked with a friend on Color Jumper, a mobile game project using Godot and GitHub

Catapult Analysis, VEX Robotics 2024-2025
• Wrote a report on optimizing a VEX Robotics catapult system through theoretical and experimental analysis

LEADERSHIP & SERVICE

Eagle Scout, Boy Scouts of America 2024
• Eagle Scout project: built two map/bulletin display cases for a local high school; 70 personal hours, 172 total hours
• Troop roles included Senior Patrol Leader, Patrol Leader, and Scribe

Robotics Camp, Roseville, CA 2024
• Helped plan and run a free 5-day camp for 26 elementary and middle school students
• Prepared demonstrations, lesson materials, and hands-on robotics/coding activities with teammates

Robotics Club, Oakmont High School 2022-2024
• Team captain 2022-2024; vice president 2023-2024
• Worked on VEX robot design and programming;

HONORS & ACTIVITIES

High School Valedictorian | National Merit Scholarship Finalist | IB Diploma (42/45) | Eagle Scout | Seal of Bilingualism (French/English)
Coding competitions: 4th non-university ACPC, 3rd HTPE Code Wars